```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
from scipy import stats
```

```
df=pd.read_csv('aerofit_treadmill.csv')
df.head()
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

Next steps: Generate code with df View recommended plots New interactive sheet

```
df.shape
```

```
(180, 9)
```

```
# prompt: df datatype

df.dtypes
```

	0
Product	object
Age	int64
Gender	object
Education	int64
MaritalStatus	object
Usage	int64
Fitness	int64
Income	int64
Miles	int64

dtype: object

```
df.Product.unique()
```

```
array(['KP281', 'KP481', 'KP781'], dtype=object)
```

```
df.isnull().sum()
```

	0
Product	0
Age	0
Gender	0
Education	0
MaritalStatus	0
Usage	0
Fitness	0
Income	0
Miles	0

dtype: int64

So , We can see that in total we have Three and we want to find Which

product fit for which type of audience

```
con_variable=df[['Age', 'Education', 'Income', 'Fitness', 'Miles']]
con_variable.head()
```

	Age	Education	Income	Fitness	Miles
0	18	14	29562	4	112
1	19	15	31836	3	75
2	19	14	30699	3	66
3	19	12	32973	3	85
4	20	13	35247	2	47

Next steps: Generate code with con_variable View recommended plots New interactive sheet

```
con_variable.describe()
```

	Age	Education	Income	Fitness	Miles
count	180.000000	180.000000	180.000000	180.000000	180.000000
mean	28.788889	15.572222	53719.577778	3.311111	103.194444
std	6.943498	1.617055	16506.684226	0.958869	51.863605
min	18.000000	12.000000	29562.000000	1.000000	21.000000
25%	24.000000	14.000000	44058.750000	3.000000	66.000000
50%	26.000000	16.000000	50596.500000	3.000000	94.000000
75%	33.000000	16.000000	58668.000000	4.000000	114.750000
max	50.000000	21.000000	104581.000000	5.000000	360.000000

```
# prompt: con_variable count the elements

con_variable.count()
```

	0
Age	180
Education	180
Income	180
Fitness	180
Miles	180

dtype: int64

```
cat_variable=df[['Product','Gender','MaritalStatus']]
cat_variable.head()
```

	Product	Gender	MaritalStatus
0	KP281	Male	Single
1	KP281	Male	Single
2	KP281	Female	Partnered
3	KP281	Male	Single
4	KP281	Male	Partnered

Next steps: Generate code with cat_variable View recommended plots New interactive sheet

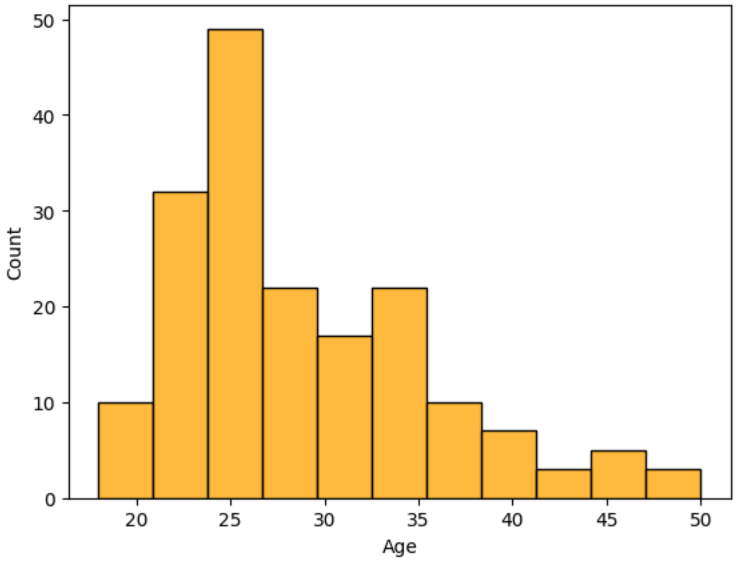
```
cat_variable.count()
```

	0
Product	180
Gender	180
MaritalStatus	180

dtype: int64

```
sns.histplot(df['Age'],color='orange')
```

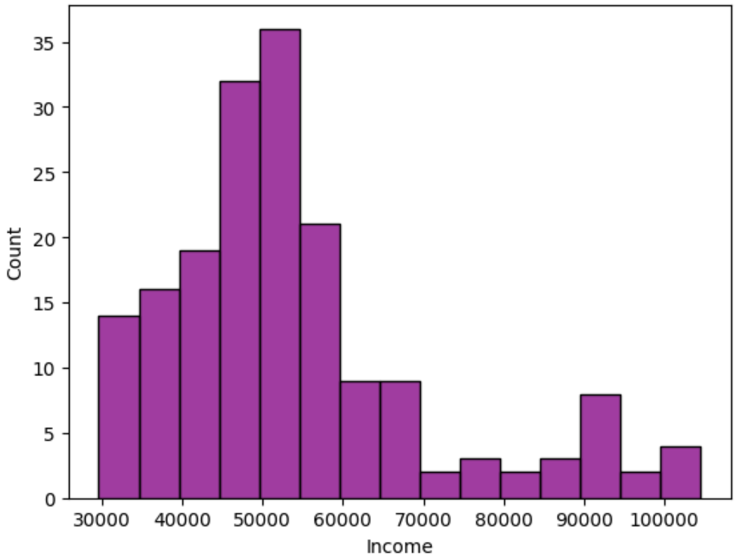
```
<Axes: xlabel='Age', ylabel='Count'>
```



In General ,it is seen that the age group between 22 to 30 are those who purchase Aerofit product

```
sns.histplot(df['Income'],color='purple')
```

```
<Axes: xlabel='Income', ylabel='Count'>
```



In General ,it is seen with this Histplot that the Income group between $45,000 to 60,000 are those who purchase Aerofit product

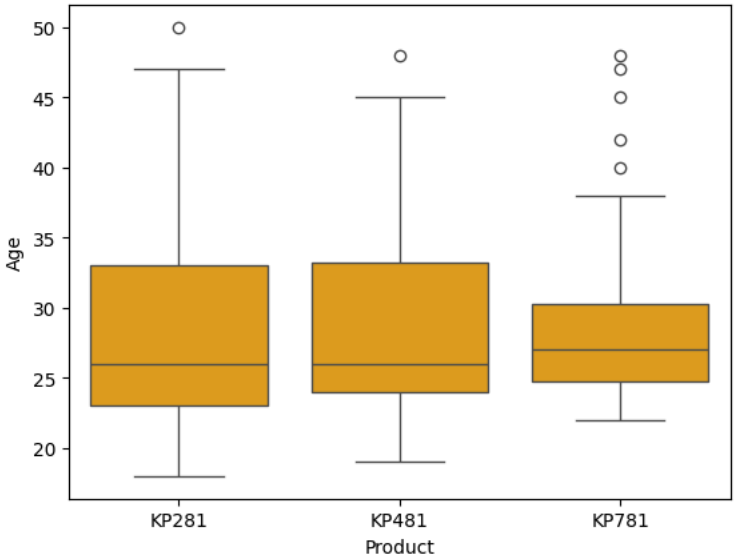
```
pa_mean=df.groupby('Product')['Age'].mean()
pa_mean
```

	Age
Product	
KP281	28.55
KP481	28.90
KP781	29.10

dtype: float64

```
sns.boxplot(x='Product',y='Age',data=df,color='orange')
```

```
<Axes: xlabel='Product', ylabel='Age'>
```



All the products have a mean age of around 28 to 29 years. However, the box plot shows that KP781 has a narrower spread, with the 75th percentile of the data around 32 years. It also has several outliers affecting the mean, so it would be better to focus on the median for a clearer picture.

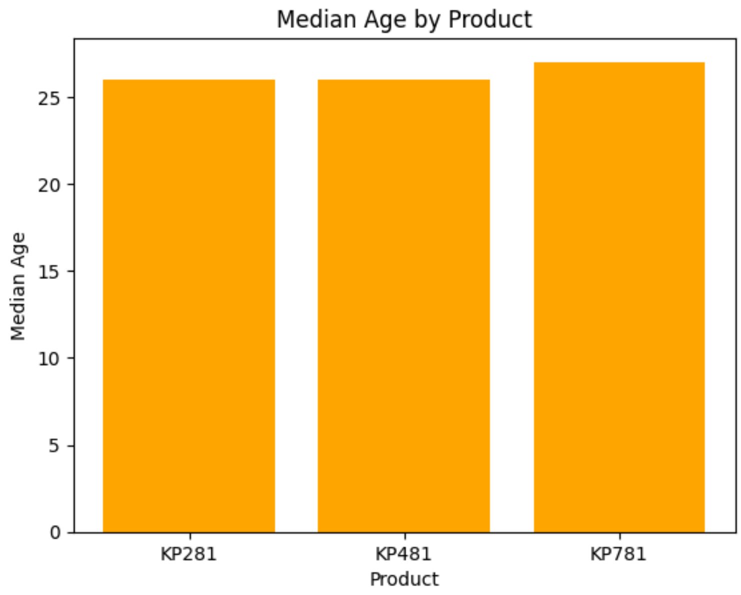
Suggestions:

- For products like **KP281** and **KP481**, the target audience could include younger customers, such as college students or those who have just started working. The mean age for these products is around 28 years, but this could be increased by focusing on middle-aged customers through targeted ads and campaigns.
- For **KP781**, which is more expensive, the mean age is also around 29 years, but this needs to increase, ideally with the upper age limit reaching over 50 years. Targeted strategies like marketing at business events or venues where middle-aged customers gather could help. Additionally, using role models or figures that resonate with this demographic could strengthen the product's appeal.

```
prod_age_median = df.groupby('Product')['Age'].median()
print(prod_age_median)
```

```
Product
KP281    26.0
KP481    26.0
KP781    27.0
Name: Age, dtype: float64
```

```
plt.bar(prod_age_median.index, prod_age_median.values,color='orange')
plt.xlabel('Product')
plt.ylabel('Median Age')
plt.title('Median Age by Product')
plt.show()
```

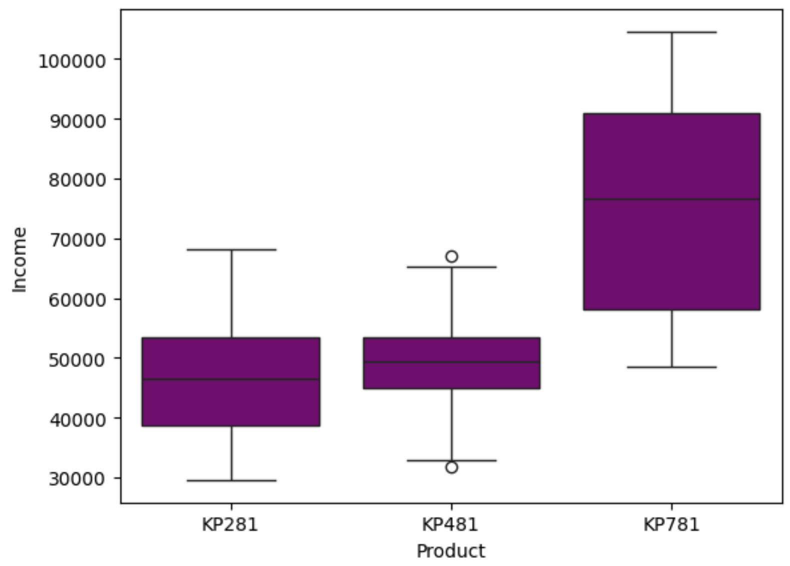


So median suggest that the KP781 median customer age is slightly higher than the rest 2 products

Product and Income Analysis

```
sns.boxplot(x='Product',y='Income',data=df,color='purple')
```

```
<Axes: xlabel='Product', ylabel='Income'>
```



In the boxplot, it's clear that customers who purchased the KP781 have significantly higher incomes compared to those who bought KP281 and KP481.

Suggestions:

- For **KP481**, the income range is quite narrow, so efforts should focus on expanding this bracket by targeting a broader audience with higher income levels.
- For **KP281**, the target should be customers with lower incomes, such as early career starters, college students, or individuals with a more limited budget. Tailored marketing towards these groups can help increase sales in this segment.

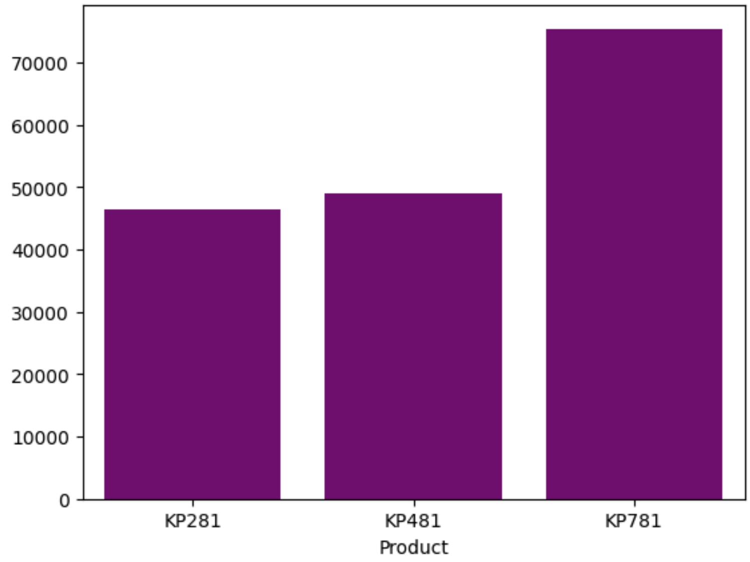
```
prod_income=df.groupby('Product')['Income'].mean()
prod_income
```

	Income
Product	
KP281	46418.025
KP481	48973.650
KP781	75441.575

dtype: float64

```
sns.barplot(x=prod_income.index,y=prod_income.values,color='purple')
```

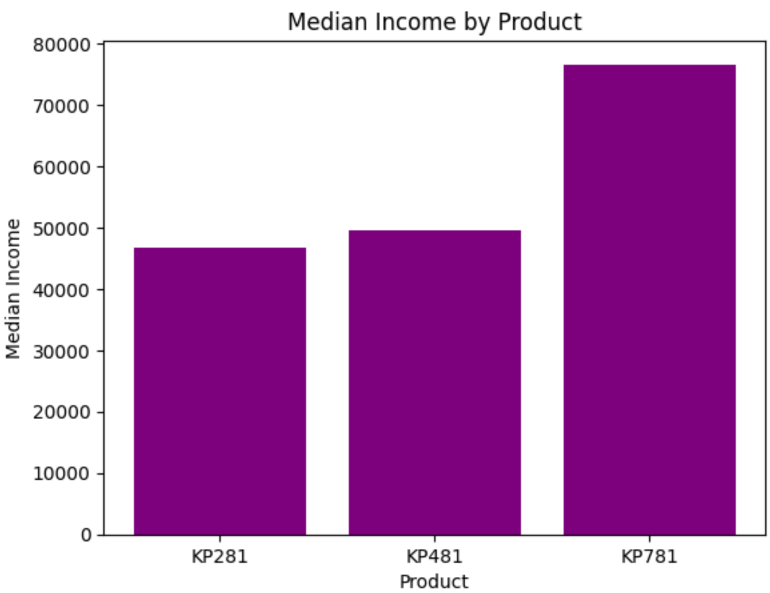
```
<Axes: xlabel='Product'>
```




```
prod_Income_median = df.groupby('Product')['Income'].median()
print(prod_Income_median)
```

```
Product
KP281    46617.0
KP481    49459.5
KP781    76568.5
Name: Income, dtype: float64
```

```
plt.bar(prod_Income_median.index, prod_Income_median.values,color='purple')
plt.xlabel('Product')
plt.ylabel('Median Income')
plt.title('Median Income by Product')
plt.show()
```



Median also suggest similar result as mean which is KP781 is higher than other two products

Fitness Analysis

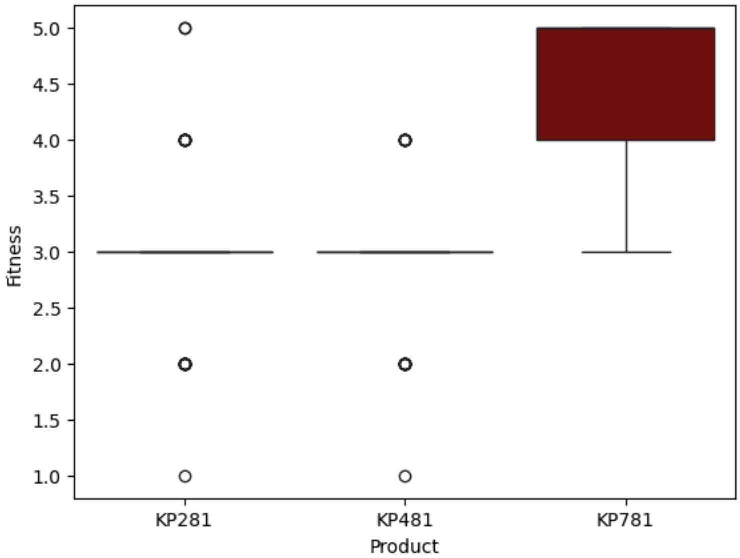
```
df.groupby('Product')['Fitness'].mean()
```

Product	Fitness
KP281	2.9625
KP481	2.9000
KP781	4.6250

dtype: float64

```
sns.boxplot(x='Product',y='Fitness',data=df,color='maroon')
```

<Axes: xlabel='Product', ylabel='Fitness'>



The data shows that the fitness level spread for KP281 and KP481 is narrow, primarily around 3.0, while for KP781, it ranges between 4.0 and 5.0, with a mean of 4.62. This suggests that customers with higher fitness levels prefer the KP781, indicating a mismatch in product placement —entry-level treadmills should be targeted at those just beginning their fitness journey.

Suggestions:

- **Marketing** is key to properly aligning product placement with customer needs. KP281 should be marketed as the ideal choice for beginners starting their fitness journey, while KP481 and KP781 should be positioned as upgrade options for more experienced users.
- To reach the right audience, leveraging role models, ads, and campaigns—such as marathons or corporate events—can effectively target customers with varying fitness aspirations.

Miles Analysis

```
df.groupby('Product')['Miles'].mean()
```

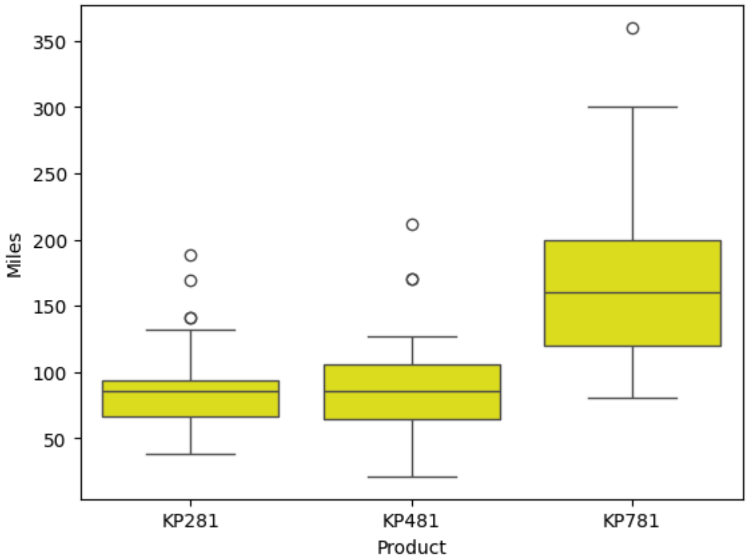
Product	Miles
KP281	82.787500
KP481	87.933333
KP781	166.900000

dtype: float64

This shows that the KP781 treadmill is ideal for users with a higher average walking/running capacity compared to the other two models. The KP281 is best suited for beginners, while the KP481 is designed for intermediate users.

```
sns.boxplot(x='Product',y='Miles',data=df,color='yellow')
```

```
<Axes: xlabel='Product', ylabel='Miles'>
```

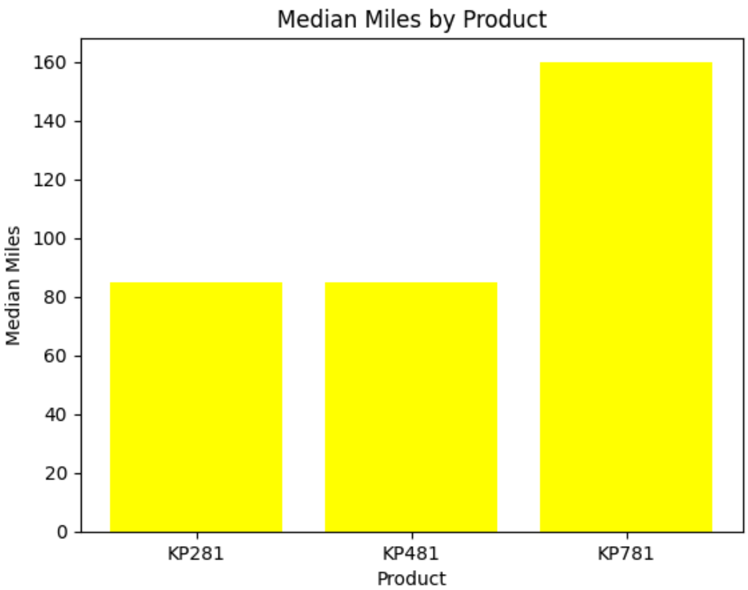


Analyzing this box plot, we observe that the data spread is widest for the KP781, but its starting point is higher than the other two products, beginning above 100 miles. Excluding outliers, both of the other products have distances below 100 miles.

```
prod_miles_median = df.groupby('Product')['Miles'].median()
print(prod_miles_median)

plt.bar(prod_miles_median.index, prod_miles_median.values,color='yellow')
plt.xlabel('Product')
plt.ylabel('Median Miles')
plt.title('Median Miles by Product')
plt.show()
```

```
Product
KP281     85.0
KP481     85.0
KP781    160.0
Name: Miles, dtype: float64
```



Gender Based Product analysis

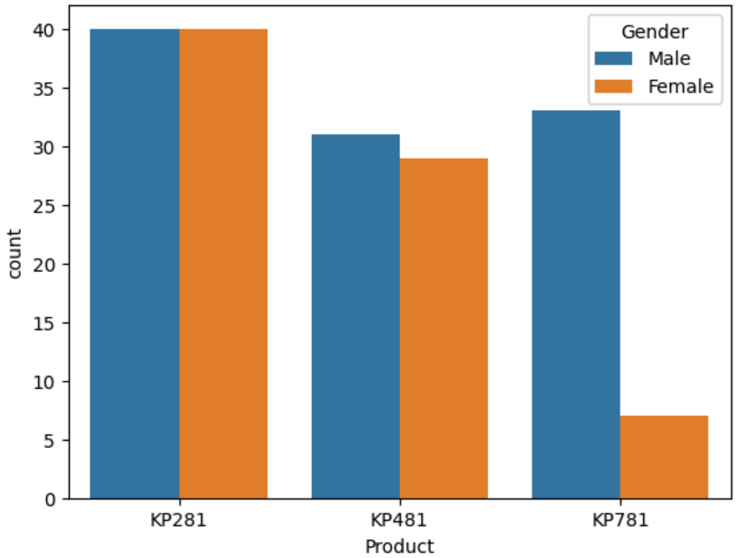
```
df.groupby('Gender')['Product'].value_counts()
```

Gender	Product	count
Female	KP281	40
	KP481	29
	KP781	7
Male	KP281	40
	KP781	33
	KP481	31

dtype: int64

```
sns.countplot(x='Product',hue='Gender',data=df)
```

```
<Axes: xlabel='Product', ylabel='count'>
```



The plot shows that KP281 has an equal number of male and female customers, while KP481 has slightly more male buyers but also a good number of female customers. KP781, however, is predominantly purchased by men.

Suggestion:

- To increase sales for KP781, the company could focus on targeting female customers through specialized ads or campaigns designed specifically for women. This approach could help broaden the product's appeal and capture a larger segment of the market.

```
# Marginal probability
pd.crosstab(df['Product'], df['Gender'], normalize='all')
```

Gender Product	Female	Male
KP281	0.222222	0.222222
KP481	0.161111	0.172222
KP781	0.038889	0.183333

The data shows the proportion of male and female customers for each product (KP281, KP481, KP781), normalized across all customers.

Insights:

1. **KP281**:
   - **22.22%** of the total customers for KP281 are female, and **22.22%** are male.
   - **Insight**: KP281 has a balanced gender distribution, appealing equally to both male and female customers. This suggests that it is a versatile product with broad appeal across genders.

2. **KP481**:
    - **16.11%** of the total customers are female, while **17.22%** are male.
    - **Insight**: KP481 has slightly more male customers than female, but the difference is not significant. This product has a broader appeal but skews somewhat towards males.
3. **KP781**:
    - Only **3.89%** of the total customers are female, while **18.33%** are male.
    - **Insight**: KP781 is predominantly purchased by male customers, with a much smaller proportion of female buyers. This suggests that KP781 either has features that appeal more to men or that marketing efforts have primarily targeted men.

Overall Insight:

- **KP281** is equally popular among both male and female customers.
- **KP481** is slightly more popular among men but still has a good representation of female customers.
- **KP781** is heavily male-dominated, indicating a need to either adjust marketing strategies or product features to better appeal to women.

Suggestions:

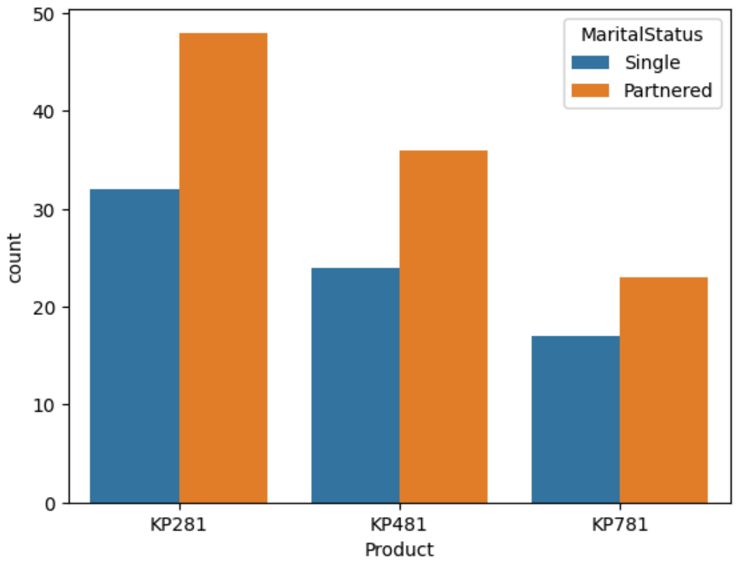
- For **KP281**, continue with inclusive marketing that targets both men and women.
- For **KP481**, maintain gender-neutral messaging but consider campaigns that appeal more to women to balance the gender distribution.
- For **KP781**, there is a significant opportunity to increase sales by targeting women. Specialized ads or campaigns focusing on women's fitness, featuring female athletes or influencers, could help bridge the gap. Highlighting features that appeal to women, such as design, functionality, or health benefits, may help increase female interest in KP781.

```
pd.crosstab(df['Product'],df['MaritalStatus'])
```

MaritalStatus	Partnered	Single
Product		
KP281	48	32
KP481	36	24
KP781	23	17

```
sns.countplot(x='Product',hue='MaritalStatus',data=df)
```

```
<Axes: xlabel='Product', ylabel='count'>
```



The trend indicates that partnered customers are purchasing more of all three treadmill products compared to single customers.

Suggestions:

- To boost sales among single customers, targeted ads should focus on promoting the benefits of fitness for individuals, such as personal health, self-improvement, and stress relief. Campaigns could emphasize flexibility, independence, and convenience—attributes that resonate with single buyers.
- Offering promotions such as **solo fitness challenges** or **discounts for first-time buyers** could also attract more single customers. Additionally, using influencers or role models who represent independent lifestyles could help connect with this audience.

```
pd.crosstab(df['Product'],df['MaritalStatus'],normalize='all')
```

MaritalStatus Product	Partnered	Single
KP281	0.266667	0.177778
KP481	0.200000	0.133333
KP781	0.127778	0.094444

The output from the `pd.crosstab()` shows the proportion of customers who are **Partnered** or **Single** for each treadmill product, normalized over all customers. Here's the insight:

Insight:

1. **KP281**:
   - 26.67% of total customers are partnered.
   - 17.78% of total customers are single.
   - **Insight**: KP281 is more popular among partnered customers, but a significant portion of buyers are single, making it appealing to both groups.
2. **KP481**:
   - 20% of total customers are partnered.
   - 13.33% of total customers are single.
   - **Insight**: KP481 also shows a higher proportion of partnered customers compared to single ones, but the gap is slightly smaller compared to KP281.
3. **KP781**:
   - 12.78% of total customers are partnered.
   - 9.44% of total customers are single.
   - **Insight**: KP781 has the lowest proportions for both partnered and single customers compared to KP281 and KP481, but the trend of more partnered customers remains consistent.

Overall Insight:

Across all products, **partnered customers** consistently purchase more treadmills than single customers, with the KP281 having the largest gap between the two. This suggests that partnered customers are more likely to invest in fitness equipment, possibly due to shared household decisions or family-oriented fitness goals.

Suggestions:

- To increase sales among **single customers**, targeted ads focusing on **individual fitness goals** and **personal empowerment** could resonate well, particularly for KP781, which has the smallest proportion of single customers.
- Highlighting the **convenience** and **benefits of home fitness** for single lifestyles may help shift the balance.

```
conditional_prob = pd.crosstab(df['MaritalStatus'], df['Product'], normalize='columns')

print(conditional_prob)
```

```
Product        KP281  KP481  KP781
MaritalStatus
Partnered        0.6    0.6  0.575
Single           0.4    0.4  0.425
```

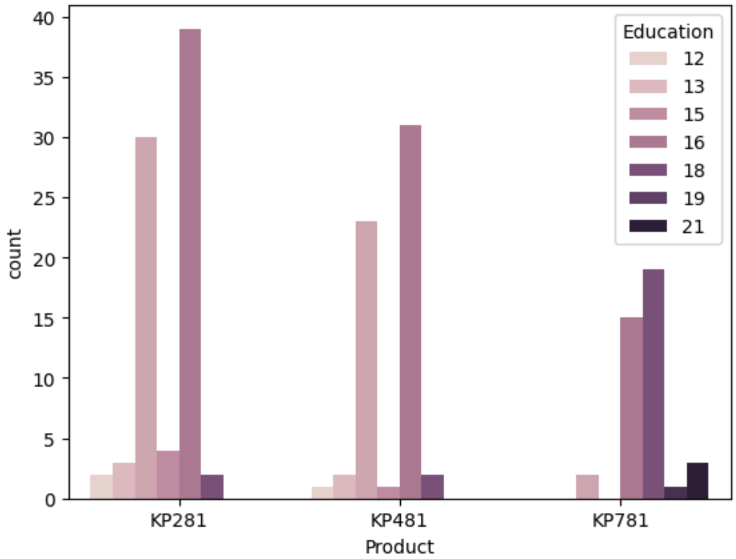
```
pd.crosstab(df['Product'],df['Education'])
```

Education Product	12	13	14	15	16	18	20	21
KP281	2	3	30	4	39	2	0	0
KP481	1	2	23	1	31	2	0	0
KP781	0	0	2	0	15	19	1	3

Double-click (or enter) to edit

```
sns.countplot(x='Product',hue='Education',data=df)
```

```
<Axes: xlabel='Product', ylabel='count'>
```



The provided data appears to represent a crosstab of the "**Product**" (KP281, KP481, KP781) against different "**Education**" levels (12, 13, 14, etc.), with the values indicating the number of customers for each product at different education levels. Here's a breakdown of the insights:

Insights:

1. **KP281**:
    - Most buyers have education levels 14, 15, and 16, with the highest number (39) at level 16.
    - Very few customers fall in the lower education levels (12 and 13) and none in higher levels (20 and 21).
    - **Insight**: KP281 seems to be most popular among customers with moderate education levels (high school and early college), likely indicating that this product appeals to people earlier in their careers or education.
2. **KP481**:
    - Similar to KP281, most customers are concentrated at education levels 14, 15, and 16, with the highest number (31) at level 16.
    - There are very few customers in the lower and higher education levels, similar to KP281.
    - **Insight**: KP481 also appeals to customers with mid-level education, but it has a slightly smaller concentration compared to KP281.
3. **KP781**:
    - The pattern is different here, with most customers having education levels 16 and above.
    - A large number of customers (19) fall at education level 18, and there are even some customers with education levels 20 and 21 (postgraduate or higher).
    - **Insight**: KP781 is more popular among highly educated customers, potentially indicating that higher education levels are associated with more disposable income or a preference for premium products.

Overall Insight:

- **KP281 and KP481** appeal more to customers with moderate education levels (14–16), while **KP781** is favored by customers with higher education levels (16 and above).
- The trend suggests that more educated customers might have the financial means or inclination to purchase the more expensive KP781 model, while KP281 and KP481 cater to a broader audience in earlier stages of their education or careers.

Suggestions:

- For **KP281 and KP481**, marketing campaigns can focus on younger customers in high school or early college stages, emphasizing affordability and accessibility.
- For **KP781**, campaigns targeting highly educated professionals or those in higher income brackets, focusing on product quality and premium features, can help capture more customers in this segment.

Median also suggest the similar result as mean

```
pd.crosstab(df['Product'],df['Education'],normalize='all')
```

Education	12	13	14	15	16	18	20	21
Product								
KP281	0.011111	0.016667	0.166667	0.022222	0.216667	0.011111	0.000000	0.000000
KP481	0.005556	0.011111	0.127778	0.005556	0.172222	0.011111	0.000000	0.000000
KP781	0.000000	0.000000	0.011111	0.000000	0.083333	0.105556	0.005556	0.016667

The provided data represents the proportions of customers in different education levels for each product (KP281, KP481, KP781), normalized over all customers. Here's the insight and breakdown:

Insights:

1. **KP281**:
   - **Education level 16** has the highest proportion of customers (21.67%), followed by **level 14** (16.67%).
   - Very small proportions of customers fall in education levels 12, 13, and 15, and none are seen in higher levels (20 and 21).
   - **Insight**: KP281 is most popular among customers with mid-level education, especially around high school and college level (education 14-16). It has almost no traction among highly educated customers.
2. **KP481**:
   - Like KP281, **education level 16** has the highest proportion of customers (17.22%), followed by **level 14** (12.78%).
   - Lower education levels (12, 13, and 15) also contribute minimally, and none are present in the higher levels (20 and 21).
   - **Insight**: KP481 shows a similar pattern to KP281, with most customers concentrated in the mid-level education brackets.
3. **KP781**:
   - The highest proportion of customers (10.56%) are at **education level 18**, with a smaller but notable proportion at **level 16** (8.33%) and **level 20** (0.56%).
   - Unlike KP281 and KP481, KP781 has customers in the higher education levels (18, 20, 21), but none at the lower levels (12, 13, 15).
   - **Insight**: KP781 attracts more highly educated customers (post-college and postgraduate levels). It's seen as a premium product that appeals to individuals with higher education and likely higher income.

Overall Insight:

- **KP281 and KP481** attract customers with mid-level education (mostly levels 14–16).
- **KP781** attracts customers with higher education (16 and above), particularly around postgraduate levels.

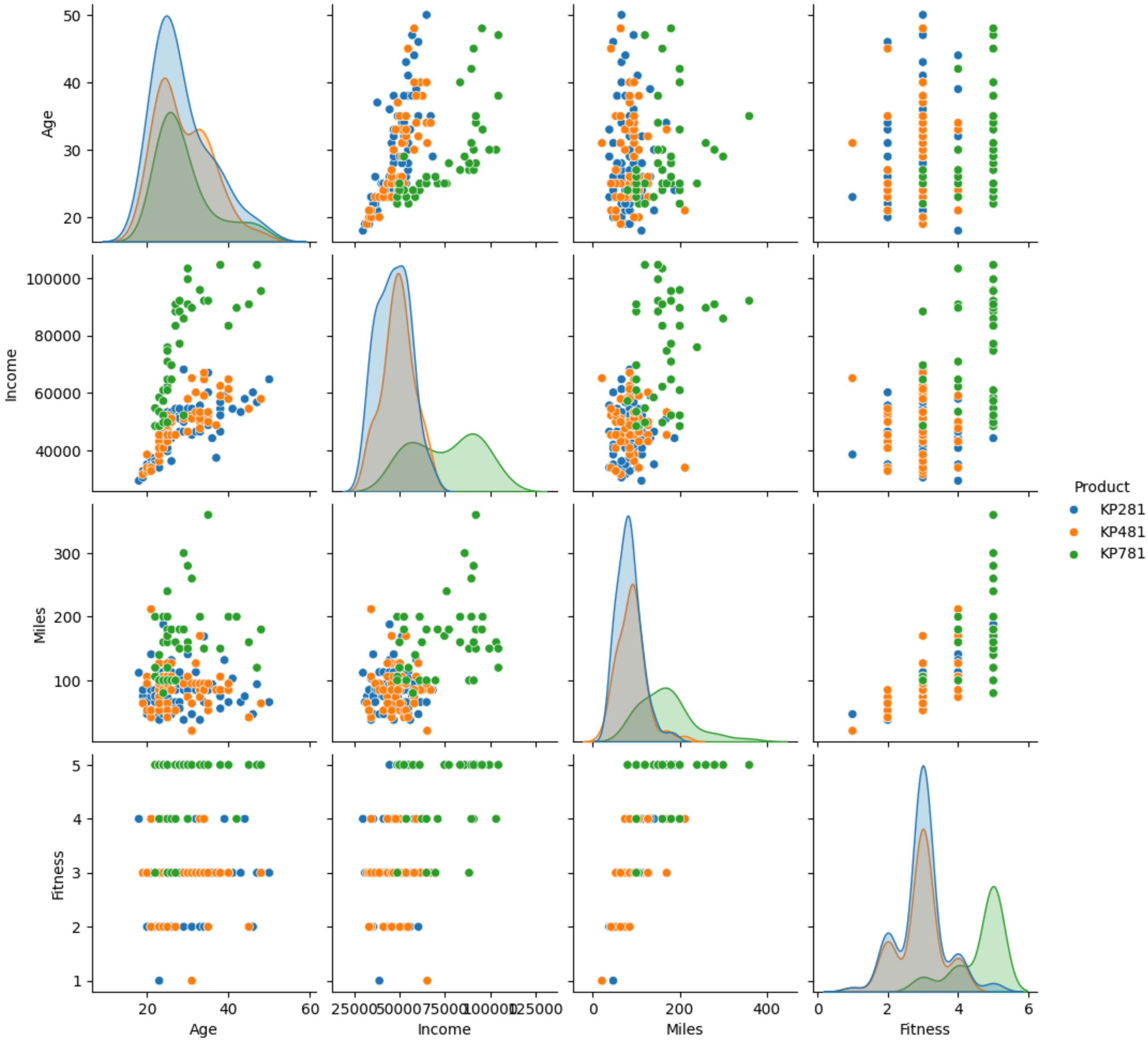
Suggestions:

- For **KP281 and KP481**, focus marketing on younger demographics or individuals in early career stages, highlighting affordability and accessibility.
- For **KP781**, target campaigns at highly educated and financially stable customers. Position it as a premium, high-performance product and use messaging that appeals to professional or educated audiences. Consider utilizing luxury or lifestyle branding to resonate with this demographic.

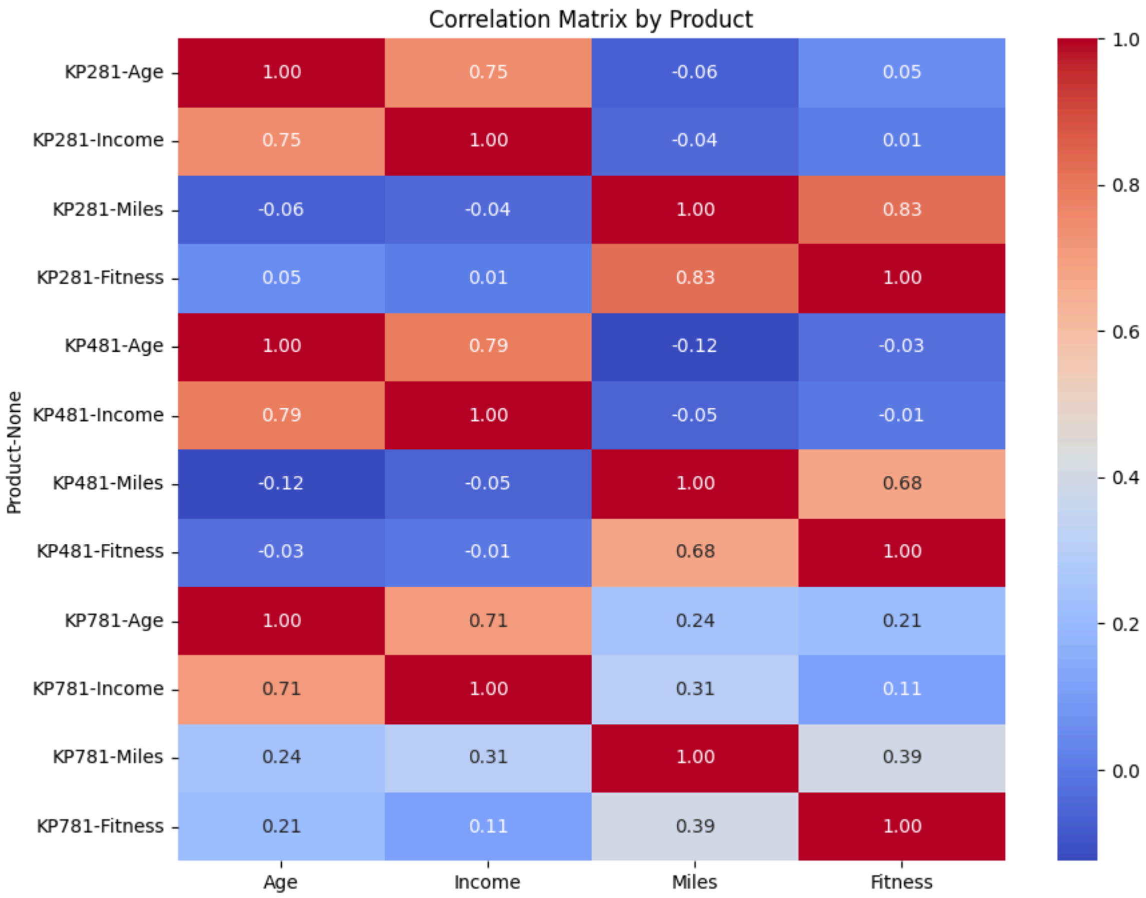
```
pd.crosstab(df['Product'],df['Education'],normalize='columns')
```

Education	12	13	14	15	16	18	20	21
Product								
KP281	0.666667	0.6	0.545455	0.8	0.458824	0.086957	0.0	0.0
KP481	0.333333	0.4	0.418182	0.2	0.364706	0.086957	0.0	0.0
KP781	0.000000	0.0	0.036364	0.0	0.176471	0.826087	1.0	1.0

```
sns.pairplot(df[['Age', 'Income', 'Miles', 'Fitness', 'Product']], hue='Product')
plt.show()
```




```
plt.figure(figsize=(10, 8))
correlation_matrix = df.groupby('Product')[['Age', 'Income', 'Miles', 'Fitness']].corr()
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Matrix by Product')
plt.show()
```



Recommendations and Actionable Insights in Simple Terms:

1. **Marketing for KP281**:
   - **What We Found**: KP281 appeals to both men and women equally.
   - **What to Do**: Keep using ads that show both men and women using the treadmill. Highlight that it's great for everyone, regardless of fitness level.
   - **How to Do It**:
     - Use diverse models in ads to represent different users.
     - Collaborate with popular fitness influencers who appeal to all genders.
     - Share posts on social media where both men and women can relate.
2. **Getting More Women to Buy KP481**:
   - **What We Found**: More men buy KP481, but there are still quite a few women.
   - **What to Do**: Focus on getting more women interested in this treadmill.
   - **How to Do It**:
     - Create ads that specifically talk to women and show them how the treadmill can help them.
     - Team up with female fitness influencers for promotions.
     - Attend events that are geared towards women, like health fairs or fun runs.
3. **Attracting Women to KP781**:
   - **What We Found**: KP781 is mostly bought by men, with very few women buying it.
   - **What to Do**: Change the marketing strategy to make it more appealing to women.
   - **How to Do It**:
     - Work with female athletes or popular women in fitness to endorse KP781.
     - Offer different designs or colors that might attract women.
     - Create ads that empower women and show how KP781 can help them reach their fitness goals.
4. **Targeting Different Age Groups**:
   - **What We Found**: KP281 and KP481 are popular with younger customers, while KP781 attracts slightly older customers.
   - **What to Do**: Tailor marketing strategies based on age.
   - **How to Do It**:

- For KP281 and KP481, focus on young adults and students. Emphasize affordability and starting a fitness journey.
- For KP781, market it as a high-end option for older, successful professionals. Highlight its premium features and health benefits.